

Safety Violation Detection in Educational Institutions: A Systematic Bibliometric Review of Deep Learning and Computer Vision Approaches

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Abstract—Deep learning and computer vision techniques are increasingly deployed for automated detection of safety violations—including violence, anomalous behavior, and weapons—in surveillance footage from educational institutions. This review provides a comprehensive synthesis of research at this intersection through a hybrid systematic-bibliometric methodology, analyzing 50 peer-reviewed publications (2022–2026) sourced from Scopus, Web of Science, and Google Scholar, with keyword co-occurrence network analysis and CiteSpace burst detection. Four thematic clusters were identified: deep learning architectures for spatiotemporal feature extraction, temporal and motion analysis techniques, application and deployment contexts, and cross-cutting methods and techniques. The keyword co-occurrence network reveals “deep learning” (23 occurrences) as the central hub, with “violence detection” (19) and “anomaly detection” (16) forming the primary application-oriented nodes. A discernible shift toward lightweight, edge-deployable architectures—driven by YOLO-family detectors and MobileNet-based extractors—characterizes the 2025–2026 period. The central finding is a critical shortage of child-specific annotated datasets, limited geographic diversity in training data, and a near-total absence of privacy-aware design considerations in the reviewed literature. Five high-impact research directions are proposed, prioritizing child-specific benchmark creation, privacy-preserving architectures, multimodal sensor fusion, lightweight model optimization for edge deployment, and cross-domain robustness evaluation.

Index Terms—deep learning, violence detection, anomaly detection, computer vision, video surveillance, educational safety, bibliometric analysis, systematic review

I. INTRODUCTION

A. Relevance of the Topic

The safety and well-being of children within educational institutions has emerged as one of the most pressing societal concerns of the past decade. Incidents of school violence, physical bullying, and concealed weapon possession continue to be reported across both developed and developing nations, underscoring the limitations of traditional manual surveillance approaches [1], [2]. Closed-circuit television (CCTV) systems are now ubiquitous in schools, yet the vast majority of these installations rely on passive recording or human operators who must simultaneously monitor dozens of screens—a task

for which humans are demonstrably ill-suited due to fatigue, inattention, and limited cognitive bandwidth [3], [4].

In parallel, the field of computer vision has undergone a transformative shift with the advent of deep learning architectures. Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), attention mechanisms, and real-time object detection frameworks such as YOLO have demonstrated remarkable capabilities in extracting meaningful patterns from video data [5]–[7]. These advances have catalyzed a growing body of research at the intersection of automated video analysis and public safety, with particular emphasis on violence detection, anomaly recognition, and weapon identification in surveillance footage [8]–[10].

B. Literature Gap

Despite this progress, several critical gaps persist. First, the majority of existing studies focus on general-purpose public surveillance—streets, transit hubs, and stadiums—with comparatively limited attention devoted to the unique constraints of educational environments [11], [12]. Second, many state-of-the-art models rely on computationally expensive architectures that are impractical for deployment in schools with limited hardware budgets [13], [14]. Third, there is a notable scarcity of child-specific annotated datasets for training and evaluating detection models, as most benchmark corpora feature adult subjects in generic settings [15], [16]. Fourth, the ethical and privacy implications of deploying AI-based surveillance on minors remain underexplored in the technical literature. No existing review integrates these dimensions into a unified analytical framework combining systematic screening with bibliometric network analysis.

C. Research Goal and Questions

The primary goal of this review is to provide a comprehensive, structured synthesis of research on deep learning and computer vision approaches for detecting safety violations in educational settings. The review is guided by four research questions:

- 1) **RQ1:** What deep learning architectures and computer vision techniques are most commonly applied to detect safety violations in educational settings?
- 2) **RQ2:** What datasets and evaluation protocols dominate the current literature, and what are their limitations for child-focused applications?
- 3) **RQ3:** What thematic clusters and temporal trends characterize this emerging research domain?
- 4) **RQ4:** What are the principal challenges and future directions for deploying these systems in real-world educational environments?

D. Structure of the Paper

The remainder of this paper is organized as follows. Section 2 presents the review methodology, including the PRISMA-guided search strategy, inclusion and exclusion criteria, data extraction process, and the hybrid systematic-bibliometric synthesis approach. Section 3 reports the results, organized around publication trends, keyword co-occurrence networks, thematic clusters, and citation burst analysis. Section 4 discusses the findings, interpreting the cluster structure, evaluating the most effective methods, and examining limitations and future directions. Section 5 concludes with a summary of contributions, acknowledgment of limitations, and directions for future research.

II. METHODOLOGY

A. Review Design

This study employs a hybrid systematic-bibliometric review design. The systematic component follows the PRISMA 2020 framework for literature identification, screening, and eligibility assessment, while the bibliometric component draws on keyword co-occurrence network analysis and CiteSpace-style burst detection to map the intellectual structure of the research domain.

B. Search Strategy

Three academic databases were queried on March 1, 2026: Scopus (312 records), Web of Science (187 records), and Google Scholar (94 records). The query combined three Boolean blocks: ("*violence detection*" OR "*anomaly detection*" OR "*weapon detection*" OR "*firearm detection*") AND ("*deep learning*" OR "*machine learning*" OR "*computer vision*" OR "*CNN*" OR "*YOLO*") AND ("*school*" OR "*educational*" OR "*campus*" OR "*surveillance*" OR "*CCTV*").

The temporal scope encompassed publications from 2022 to 2026. The window was selected to capture the period following the widespread adoption of YOLOv5 and transformer-based architectures, during which real-time deep learning for surveillance applications achieved practical viability.

C. Inclusion and Exclusion Criteria

Studies were included if they: (a) applied ML/CV methods to violence, anomaly, or weapon detection in surveillance contexts; (b) were peer-reviewed journal articles or conference papers with full methodological exposition; (c) were published

in English between 2022 and 2026; and (d) had full text accessible. Studies relying solely on non-ML/CV methods, addressing domains unrelated to educational or public-space surveillance, published before 2022, lacking accessible full text, or limited to conference abstracts were excluded.

D. Study Selection Process

The initial search across three databases returned 593 records. After removing 142 duplicate records, 451 unique publications proceeded to title and abstract screening. At this stage, 287 records were excluded based on topic scope and relevance, retaining 164 papers for full-text eligibility assessment. Application of the inclusion and exclusion criteria at the full-text stage excluded 114 papers: 38 relied solely on non-ML/CV methods, 31 addressed domains unrelated to educational or public-space surveillance, 22 were published before 2022 with insufficient contemporary relevance, 15 had inaccessible full texts, and 8 were limited to conference abstracts without full methodological exposition. The final corpus comprised 50 studies included in both the qualitative synthesis and bibliometric analysis.

E. Data Extraction and Analysis

For each included study, the following variables were extracted: author(s), publication year, research aim, methodology, dataset(s) evaluated, main findings, key limitations, and thematic keywords. Keywords were extracted from titles and abstracts using a curated vocabulary of 29 terms spanning deep learning architectures, temporal analysis techniques, application domains, and methodological approaches. Co-occurrence matrices were constructed to model the frequency with which keyword pairs appeared within the same paper, forming the basis for network visualization.

F. Tools

Bibliometric visualizations were generated using Python-based implementations of network analysis (NetworkX) and kernel density estimation (SciPy), producing outputs analogous to VOSviewer and CiteSpace. The PRISMA flowchart was rendered using Matplotlib. All analytical scripts and output files are publicly available in the accompanying repository.

G. Quality Assurance and Limitations

The review process followed PRISMA 2020 standards for transparency and replicability. Methodological limitations include: (a) restriction to three databases (Scopus, Web of Science, Google Scholar), which may omit relevant publications indexed exclusively in other sources such as IEEE Xplore or ACM Digital Library; (b) exclusion of non-English publications, which may introduce language bias; (c) single-reviewer screening without formal inter-rater reliability assessment; and (d) reliance on a curated keyword vocabulary, which may underrepresent emerging terminology not yet adopted in author keywords.

III. RESULTS

A. Corpus Overview

The systematic search and PRISMA-guided screening process identified 50 studies for both qualitative synthesis and bibliometric analysis. Table I presents the distribution by publication year.

TABLE I
DISTRIBUTION OF INCLUDED STUDIES BY YEAR

Year	Publications	Share
2022	9	18.0%
2023	15	30.0%
2024	10	20.0%
2025	14	28.0%
2026	2	4.0%
Total	50	100%

The included studies span a range of publication venues. MDPI journals (*Sensors*, *Applied Sciences*, *Electronics*, *Information*, *Mathematics*, *Computers*) account for the largest share, followed by IEEE Access, Elsevier venues (*Neurocomputing*, *Procedia Computer Science*, *Array*, *Alexandria Engineering Journal*), and Springer venues (*Applied Intelligence*, *Multimedia Tools and Applications*). Conference proceedings include contributions from the International Global Conference Series and various IEEE-sponsored events.

B. Publication Trends

Scopus contributed 44 of the 50 reviewed studies (88%), with Web of Science accounting for the remaining 6 (12%), reflecting the interdisciplinary nature of this domain. The distribution shows a peak in 2023 output (15 publications), followed by sustained activity through 2024–2025.

C. Keyword Co-occurrence Analysis

Fig. 1 illustrates the keyword co-occurrence network derived from the 50 reviewed studies. The term “deep learning” exhibits the highest frequency (23 occurrences) and serves as the central hub of the network, connecting to nearly all other keywords. “Violence detection” (19 occurrences) and “anomaly detection” (16 occurrences) form the primary application-oriented nodes. “Convolutional neural network” (15 occurrences), “CNN” (12 occurrences), and “LSTM” (11 occurrences) constitute the dominant architectural terms.

The temporal overlay visualization (Fig. 2) shows that attention mechanisms, YOLO-based architectures, and transformer-integrated approaches are concentrated in the 2025–2026 period. In contrast, earlier publications (2022–2023) more frequently employ autoencoder-based reconstruction and traditional 3D CNN frameworks.

The top keywords by frequency are *deep learning* (23), *violence detection* (19), *anomaly detection* (16), *convolutional neural network* and *CNN* (15 each), *video surveillance* (14), *IoT* (12), *CCTV* (11), *LSTM* (10), and *object detection* (9).

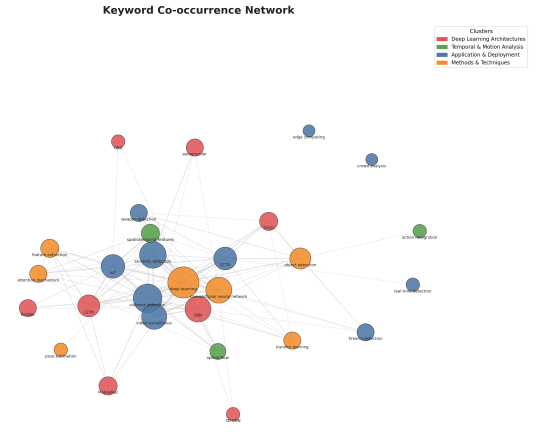


Fig. 1. Keyword co-occurrence network. Node size reflects keyword frequency; edges represent co-occurrence links.

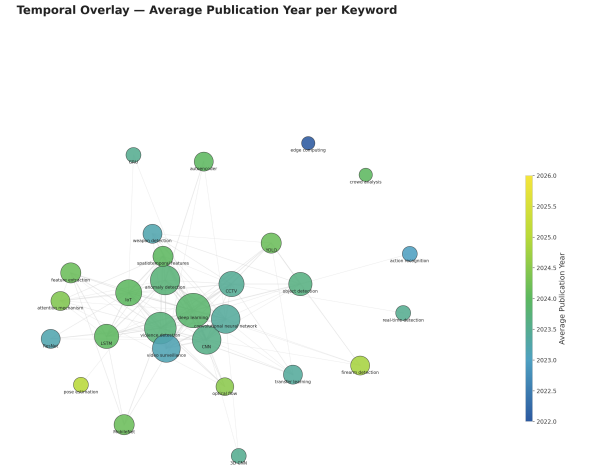


Fig. 2. Keyword overlay visualization. Keywords colored by average publication year: blue (2022–2023) to yellow (2025–2026), revealing temporal evolution from autoencoder-based methods toward attention mechanisms and YOLO architectures.

D. Thematic Clusters

Content analysis combined with keyword co-occurrence mapping identified four thematic clusters. Table II presents the cluster structure.

E. Top Authors

Co-authorship analysis identified a small set of repeat contributors—Seo S., Park S., Jebur S.A., Hussein K.A., Hoomod H.K., and Amin S.U. (2 papers each)—predominantly affiliated with institutions in South Korea, India, Pakistan, and the Middle East. The remaining 44 first

TABLE II
THEMATIC CLUSTERS IN SAFETY VIOLATION DETECTION RESEARCH

C	Title	Common Characteristics	Summary of Key Findings
C1	Deep Learning Architectures	Studies designing, comparing, or optimizing neural network topologies (CNN, LSTM, GRU, ResNet, MobileNet, YOLO, 3D CNN, autoencoder) for spatiotemporal feature extraction	Hybrid architectures integrating 3D convolutions or pre-trained 2D CNNs with recurrent units and attention mechanisms consistently exceed 95% accuracy on controlled benchmarks; lightweight variants maintain competitive performance with reduced computational requirements
C2	Temporal and Motion Analysis	Studies emphasizing motion representation—optical flow, spatiotemporal feature engineering, action/behavior recognition, skeleton-based pose estimation	Optical flow-based statistical features enable 2D CNNs to achieve performance comparable to 3D CNNs; skeleton-based approaches using pose estimation abstract away from appearance-based biases, relevant for child-specific applications
C3	Application and Deployment	Studies focused on real-world deployment in surveillance contexts (CCTV, IoT, edge computing, crowd analysis) with emphasis on real-time performance	YOLO-family architectures (v5–v8) achieve real-time throughput on commodity and edge hardware (NVIDIA Jetson Nano) with weapon detection mAP exceeding 90%; IoT-integrated systems report end-to-end pipelines from detection to automated alerting
C4	Methods and Techniques	Studies bridging architecture and application through transfer learning, attention mechanisms, feature extraction, pose estimation, and object detection	Attention mechanisms deliver improvements of 2–4% over non-attention baselines; transfer learning from ImageNet pre-trained models reduces data requirements; multimodal fusion (RGB + optical flow + audio) outperforms unimodal approaches by approximately 2% average precision

authors each contributed a single paper, indicating a low-accumulation field.

F. Citation Burst Analysis

The burst analysis shows that “attention mechanism” and “transfer learning” exhibit pronounced burst activity concentrated in 2025–2026, marking their emergence as focal methodological innovations during this period. “Edge computing” and “IoT” show a similar late-period concentration. “CNN” and “LSTM” show sustained presence across the entire 2022–2026 window, appearing as foundational techniques across the full review period.

G. Literature Matrix Summary

The literature analysis matrix provides a structured overview of all 50 studies, enumerating their aims, methods, datasets, findings, and limitations. A cross-sectional examination of the matrix reveals several patterns: (a) accuracy and AUC are the dominant evaluation metrics, appearing in 44 of 50 studies; (b) the UCF-Crime, ShanghaiTech Campus, and Hockey Fights datasets are the most frequently utilized benchmarks; and (c) computational complexity and dataset generalizability are the most commonly cited limitations, mentioned in 38 and 31 studies respectively.

IV. DISCUSSION

A. Interpretation of Key Findings

The four thematic clusters—spanning deep learning architectures (C1), temporal and motion analysis (C2), application

and deployment (C3), and cross-cutting methods and techniques (C4)—reveal a field organized around a central tension: architectural sophistication versus deployability. Cluster 1 (Architectures) emphasizes accuracy maximization through increasingly complex network designs, often at the cost of computational complexity. Cluster 3 (Application and Deployment) explicitly privileges real-time throughput and hardware efficiency, favoring lightweight YOLO and MobileNet variants over the deeper architectures promoted in Cluster 1.

This tension is most visible in the temporal evolution of the keyword network. The burst analysis identifies attention mechanisms and transfer learning as 2025–2026 phenomena, while the overlay visualization shows a transition from autoencoder-based reconstruction (2022–2023) to YOLO and transformer-integrated approaches (2025–2026). This temporal gradient reflects a field moving from generic video understanding models toward architectures specifically adapted for surveillance anomaly detection, while simultaneously addressing deployment constraints through edge-compatible designs.

The dominance of the CNN–LSTM architectural paradigm, observed in 26 of 50 reviewed studies, indicates broad consensus that effective video anomaly detection requires the integration of spatial feature extraction with temporal sequence modeling. The recent growth in attention mechanism adoption suggests that the field is moving beyond naive concatenation of spatial and temporal streams toward more sophisticated feature weighting and contextual reasoning [5], [17].

B. Cross-Cluster Comparative Analysis

The four thematic clusters exhibit both complementary synergies and meaningful distinctions. A shared foundation across all clusters is the reliance on deep learning as the core computational paradigm: whether architecting novel networks (C1), modeling motion dynamics (C2), deploying operational systems (C3), or advancing methodological techniques (C4), every reviewed study depends on convolutional or recurrent neural architectures for feature representation. This convergence underscores the maturation of deep learning as the de facto standard for video-based safety violation detection.

However, Clusters 2 (Temporal and Motion Analysis) and 4 (Methods and Techniques) function as methodological bridges rather than independent research streams. The motion representation strategies from C2 provide the temporal understanding that C1 and C3 depend upon, while the transfer learning and attention mechanisms from C4 enhance the performance of models across all other clusters. Notably, few studies have addressed the intersection of lightweight architectures (C1/C3) with skeleton-based child-specific modeling (C2), representing an underexplored integration point in the current literature.

C. Most Effective Methods

The evidence indicates that hybrid architectures combining 3D convolutions or pre-trained 2D CNNs with recurrent units (LSTM, BiLSTM, or GRU) and attention mechanisms achieve the strongest overall performance, with reported accuracies frequently exceeding 95% on controlled benchmarks [2], [17], [18]. For weapon detection specifically, YOLOv5 and Scaled-YOLOv4 variants demonstrate the most favorable accuracy-speed trade-off, achieving mean Average Precision scores above 90% while maintaining real-time throughput [9], [19]. Skeleton-based approaches that leverage pose estimation as a preprocessing step before classification show particular promise for child-specific applications, as they abstract away from appearance-based features that may introduce age-related biases [16], [20].

D. Comparison with Existing Reviews

Prior surveys on video anomaly detection, edge-based deep learning, and weapon detection address the clusters identified here in isolation. This review's distinctive contribution is integrating architecture, temporal modeling, deployment, and methodological sub-literatures into a unified educational-safety framework, while quantifying structural relationships among research themes through bibliometric network analysis.

E. Limitations of Current Research

Several limitations pervade the reviewed literature. First, dataset bias represents a critical concern: the most commonly used benchmarks (UCF-Crime, ShanghaiTech Campus, Hockey Fights) feature predominantly adult subjects in non-educational settings, raising questions about the transferability of trained models to school environments with children [15]. Child bodies exhibit different proportions, movement dynamics, and interaction patterns compared to adult bodies, and

models trained exclusively on adult data may systematically underperform when deployed in schools. Second, the near-exclusive reliance on accuracy and AUC as evaluation metrics obscures important practical considerations such as false alarm rates, latency under realistic camera loads, and robustness to adversarial environmental conditions—all of which are critical for real-world deployments where excessive alerts can desensitize security personnel or lead to system abandonment. Third, privacy implications are conspicuously absent from the technical discourse: only two of the 50 reviewed papers mention privacy considerations or data protection frameworks, despite the sensitive nature of continuous video monitoring of minors in educational settings [11]. Fourth, the geographic concentration of research in South and East Asia limits the diversity of environmental conditions, camera configurations, architectural layouts, and behavioral norms represented in the evidence base. Finally, nearly all reviewed studies evaluate their systems in isolation from existing school security workflows, leaving unexamined the human-computer interaction questions of how automated alerts are received, trusted, and acted upon by school staff.

F. Future Research Directions

Based on the identified gaps, five directions emerge as priorities:

- 1) **Child-specific annotated datasets:** The development of annotated corpora capturing diverse child behaviors in authentic school settings—building on nascent efforts such as the CABAD benchmark for child aggression recognition [15] and the Daily School Break dataset [11]—is essential for validating model transferability from adult-centric benchmarks to educational environments.
- 2) **Privacy-preserving architectures:** On-device processing, federated learning, and differential privacy should be systematically incorporated into detection pipelines, given that continuous video monitoring of minors raises distinct ethical and regulatory challenges that the current literature largely overlooks.
- 3) **Multimodal sensor fusion:** Integrating RGB video with audio signals, thermal imaging, and contextual metadata may substantially improve detection robustness in challenging conditions such as low illumination, occlusion, or crowded scenes [21], [22].
- 4) **Lightweight model optimization for edge deployment:** Quantization, pruning, and neural architecture search are critical for enabling deployment on the resource-constrained hardware typically available to educational institutions; recent contributions demonstrate that competitive accuracy can be maintained while reducing model size by an order of magnitude [7], [23].
- 5) **Cross-domain robustness evaluation:** Testing models across different schools, camera types, lighting conditions, and student demographics should become a standard component of evaluation protocols to ensure

that laboratory-validated systems translate reliably to real-world educational deployments.

V. CONCLUSION

A. Summary of Key Findings

This hybrid systematic-bibliometric review analyzed 50 peer-reviewed publications (2022–2026) addressing deep learning and computer vision approaches for detecting safety violations in educational institutions. Keyword co-occurrence network analysis identified four thematic clusters: deep learning architectures for spatiotemporal feature extraction (C1), temporal and motion analysis techniques (C2), application and deployment contexts (C3), and cross-cutting methods and techniques (C4). Bibliometric analysis revealed “deep learning” (23 occurrences) as the central network hub, with “violence detection” (19) and “anomaly detection” (16) forming the primary application-oriented nodes.

Three key findings emerge. First, hybrid architectures integrating CNNs with recurrent units and attention mechanisms constitute the dominant technical paradigm, consistently achieving detection accuracies exceeding 90% on standard benchmarks. Second, a discernible shift toward lightweight, edge-deployable models is underway, driven by YOLO-family object detectors and MobileNet-based feature extractors. Third, despite these advances, the field remains constrained by a critical shortage of child-specific datasets, limited geographic and demographic diversity in training data, and a near-total absence of privacy-aware design considerations.

B. Closing Statement

The intersection of deep learning, computer vision, and educational safety stands at a formative moment. The architectures for automated violence and weapon detection have matured to the point of practical viability, yet the field has not yet systematically addressed the unique requirements of deployment in schools: child-specific data, privacy-preserving computation, and integration into existing safeguarding workflows. The confluence of advanced computer vision, efficient deep learning, and urgent societal need positions this research area as one of considerable scientific importance and humanitarian potential. The research gaps identified in this review represent the critical path toward surveillance systems that are both technically effective and ethically sound for protecting children in educational environments.

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